Liquid intrusion and evaporation in low-pressure cavities: numerical investigation with STAR-CCM+

Einströmung und Verdampfung einer Flüssigkeit unter Niedrigdruckbedingungen: numerische Simulationen mit STAR-CCM+

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1 Numerical Methods

1.1 Governing Equations

Multiphase flow is modelled in Star-CCM+ using various frameworks. The two-fluid - eulerian multiphase - framework solves a set of mass, momentum, and energy conservation equations for both the fluids to model two-phase flow. The local instantaneous conservation equations are euler averaged resulting in interfacial terms that define the influences of the phases upon each other.

$$\frac{\partial}{\partial t}\int_V \alpha_i\rho_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = \int_V \sum_{j\neq i}(m_{ij}-m_{ji})\,\mathrm{d}V + \int_V \mathrm{S}_i^\alpha\,\mathrm{d}V \tag{1.1a}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\mathbf{v}_i\,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\mathbf{v}_i\otimes\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = &-\int_V \alpha_i\nabla\mathrm{p}\,\mathrm{d}V + \int_V \alpha_i\rho_i\mathbf{g}\,\mathrm{d}V\\ &+\int_{\partial V}\left[\alpha_i\left(\mathbf{T}_i+\mathbf{T}_i^T\right)\right]\cdot\mathrm{d}\mathbf{a} + \int_V \mathbf{M}_i\,\mathrm{d}V + \int_V (\mathbf{F_{int}})_i\,\mathrm{d}V\\ &+\int_V \mathbf{S}_i^\alpha\,\mathrm{d}V + \int_V \sum_{i=1}^{n}(m_{ij}\mathbf{v}_j - m_{ji}\mathbf{v}_i)\,\mathrm{d}V\end{aligned} \tag{1.1b}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\mathrm{E}_i\,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\mathrm{H}_i\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = &\int_{\partial V} \alpha_i(k_{\mathrm{eff}})_i\nabla T_i\cdot\mathrm{d}\mathbf{a} + \int_{\partial V}\mathbf{T}_i\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} + \int_V \mathbf{f}_i\cdot\mathbf{v}_i\,\mathrm{d}V + \int_V \mathrm{S}_{u,i}\,\mathrm{d}V\\ &+\int_V \sum_{ij} Q_i^{(ij)}\,\mathrm{d}V + \int_V \sum_{j\neq i}(m_{ij}-m_{ji})h_i(T_{ij})\,\mathrm{d}V\end{aligned} \tag{1.1c}$$

The eulerian multiphase framework is capable of resolving dispersed and separated flow whereas the volume of fluid implementation assumes sufficient mesh resolution to resolve the interface and models separated flows. Alongside the three conservation equations, the model solves an equation for volume fraction transport.

$$\frac{\partial}{\partial t}\int_V \rho\,\mathrm{d}V + \int_{\partial V}\rho\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = \int_V \sum_i \mathrm{S}_i^\alpha\rho_i\,\mathrm{d}V \tag{1.2a}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_{\partial V}\rho\mathbf{v}\,\mathrm{d}V + \int_{\partial V}\rho\mathbf{v}\otimes\mathbf{v}\cdot\mathrm{d}\mathbf{a} = &-\int_V \nabla\mathrm{p}\,\mathrm{d}V + \int_V \rho\mathbf{g}\,\mathrm{d}V + \int_{\partial V}\left[\left(\mathbf{T}+\mathbf{T}^T\right)\right]\cdot\mathrm{d}\mathbf{a}\\ &+\int_V \mathbf{f_b}\,\mathrm{d}V - \sum_i\int_{\partial V}\alpha_i\rho_i\mathbf{v}_i\otimes\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} + \int_V \mathbf{S}_i^\alpha\,\mathrm{d}V\end{aligned} \tag{1.2b}$$

$$\frac{\partial}{\partial t}\int_V \rho\mathrm{E}\,\mathrm{d}V + \int_{\partial V}\left[\rho\mathrm{H}\mathbf{v} + \sum_i \alpha_i\rho_i\mathrm{H}_i\mathbf{v}_{d,i}\right]\cdot\mathrm{d}\mathbf{a} = -\int_{\partial V}\dot{\mathbf{q}''}\cdot\mathrm{d}\mathbf{a} + \int_{\partial V}\mathbf{T}\cdot\mathbf{v}\,\mathrm{d}\mathbf{a} + \int_V \mathbf{f_b}\cdot\mathbf{v}\,\mathrm{d}V + \int_V \mathrm{S_E}\,\mathrm{d}V \tag{1.2c}$$

$$\frac{\partial}{\partial t}\int_V \alpha_i\,\mathrm{d}V + \int_{\partial V}\alpha_i\mathbf{v}\cdot\mathrm{d}\mathbf{a} = \int_V\left(\mathrm{S}_i^\alpha - \frac{\alpha_i}{\rho_i}\frac{D\rho_i}{Dt}\right)\mathrm{d}V - \int_V \frac{1}{\rho_i}\nabla(\alpha_i\rho_i\mathbf{v}_{d,i})\,\mathrm{d}V \tag{1.2d}$$

1.2 Modelling Turbulence

Flows are generally turbulent. Turbulence is chracterised by irregular fluid motion dominated by eddies and vortices across a wide range of spatial and temporal scales. The largest eddies extract kinetic energy from the mean flow, which is then, through a cascade, transferred to the eddies at the smallest scale where the energy is dissipated due to viscosity, raising the internal energy of the fluid [1, 2].

Although turbulence can be modelled in various ways to various levels of resolutions, turbulence in this study has been modelled using the Shear Stress Transport (SST) k-w model. The SST k-w model is a two-equation

eddy-viscosity turbulence model based on the Reynolds-Averaged Navier-Stokes (RANS) equations. The model combines the advantages of the k-ω model near walls—which accurately resolves boundary layer behaviour—and the k-ε model in the free stream, enhancing robustness and accuracy in separated flows [3]. The SST approach uses a blending function to smoothly transition between these models, which is especially effective in flows with strong adverse pressure gradients and separation, such as multiphase cavity flows [Menter1994].

RANS models result from decomposing flow variables into mean and fluctuating components, applying Reynolds averaging, which filters out turbulent fluctuations below the averaging scale [4, 5]. This operation yields additional Reynolds stress terms representing momentum transport by turbulence, which require closure. In order to draw closure relations, eddy-viscosity models of turbulence assume that:

- the turbulent scalar flux vector aligns with the mean scalar gradient.
- the Reynolds stress tensor is proportional to the mean velocity strain rate through an eddy viscosity.

While these assumptions simplify modelling, they tend to be, more often than not, invalid as they do not fully capture turbulence physics, especiialy in anisotropic or non-equilibrium flow [6]. This limitation can be significant in regions with complex strain rates or strong secondary flows.

The implementation of the model in Star-CCM+ can be further controlled to adjust the realisability. Realisability is the postive semi-definiteness of the reynolds stress tensor. The specification of turbulence is done using the turbulence intensity and the length scale. The inherent anisotropy in turbulence is accounted for with quadratic and cubic constitutive options in the modelling of reynolds stresses. The parameters in the model are left on their default values as they are empirically modelled.

1.3 Evaporation

The evaporation process is distinguished between occuring at saturation temperatures or not. Evaporation at saturation temperatures, known as vaporisation, is a much faster process in comparison to evaporation at normal [7]. In case of vaporisation along a heated wall, most CFD models of vaporisation treat the energy transfer by partitioning the heat flux at the wall into the latent heat of the bubble vaporisation process and the sensible heat transferred to the liquid through conduction, convection, and quencing processes. This is determined by the area fractions covered by the liquid and vapour. Correlations between the heat flux and wall superheat [what is superheat] have been made using empirical results or based on modelling of sub-processes.

The sub-processes that are primarily taken into account in such determinations are:

- number density of nucleation sites
- diameter of bubble at time of detachment from the surface
- frequency of bubble detachment

Nucleation affects the number and distribution of bubbles in the fluid. Walls, particularly so in case they're heated, are more suspectible to be nucleation sites for bubble growth as the activation energy for bubble growth is much lower. This nucleation can be modelled through the domain as homogenous or heterogenous - where nucleation sites are modelled using a presumed PDF function or be determined by empirical correlations. [8]

After a stable bubble nucleus has been formed, the growth of the bubble becomes pertinent in the evaluation of the diameter at departure and the interphase energy and mass transfer. In the case of an ideal spherical bubble undergoing expansion in a uniformly superheated liquid, the rate of growth is determined by the surface tension, liquid inertia, and the differnce in pressures inside and outside the bubble. Surface tension dominates and restricts the initial bubble growth. This initial rapid expansion of the bubble results in the formation of a thin liquid layer separating the vapour from the wall surface. This microlayer can be a significant contributor to the bubble growth and heat transfer process. As the radius increases and the liquid layer around the bubble cools due to evaporation, a decrease in vapour pressure in seen. The dynamics are controlled by thermal diffusion and

inertia. Inertial effects, however, become less relevant as the bubble growth rate decreases and the bubble radius grows. The interphase energy exchange determines further bubble growth. [9]

1.3.1 Effects of low pressure and gravity conditions

The absence of significant gravitaional forces alters the dynamics of vaporisation - affecting natural convection, buoyancy driven phenomenon, and bubble dynamics. Under terrestrial gravity, convection currents develop under thermal gradients, thus bringing cooler liquid to the heated surface as the vapour bubble rises due to buyoancy. This density driven natural convection is affected in low gravity conditions, maintaining the thermal gradient. Bulk fluid movement is affected and heat and mass transfer processes shift from being convection dominated to being diffusion dominated. [10, 11]

Under low gravity conditions, vapour bubbles tend to remain near the heat source, growing signifcantly larger and coalescing with smaller bubbles. Prolonged bubble presence on the surface results in drying out of the surface and unsteady rise in surface temperatures.[12] [Mention some numbers] In conditions where the dominance of surface tension forces becomes more pronounced, the surfae tension gradients can induce fluid circulation. This is known as Marangoni convection and is observed to affect bubble growth and heat transfer. [13]

Reduced system pressure impacts the overall intensity of heat transfer. The onset of nucleate boiling also requires higher superheat values in low pressure conditions. Decrease in pressure leads to a lower number of active nucleation sites and increases microlayer thickness found at the bubble-wall interface. The local heat flux density from microlayer evaporation is impacted and the microlayer evaporation area and depletion time is increased. The system is also susceptible to the formation of a continuous vapour film at the surface due to coalescing bubbles unable to escape the surface, reducing the overall heat transfer. [14]

The affected heat transfer and bubble growth phenomenon further result in lower critical heat fluxes and heat transfer efficiency. [15, 16]

1.3.2 Vaporisation in a superheated liquid

Flash evaporation is the intense vaporisation triggered through depressurisation of superheated fluids.

1.4 Multiple Flow Regimes and Interface Capturing

Star-CCM+ in its implementation of the eulerian-multiphase framework utilises a dynamic flow regime topology based on the local volume fractions of the interacting phases. Based on the increasing volume fraction of the secondary phase in a computational cell, the topology transitions through three principal regimes:

- dispersed in the primary phase
- having a large scale interface with the primary phase
- continuous while having the primary phase dispersed

For each cell, Star-CCM+ uses regime-specific closure models for the interfacial [OR RATHER INTERPHASE MOMENTUM AND ENERGY TRANSFER] forces. These models are blended using weighting schemes according to the calculated proportion of each regime in the cell. [17] [Gada et al. (2015): Large Scale Interface Model for Simulating Multiphase Flows (STAR-CCM+ Implementation)]

[EDIT]While there are multiple numerical methods to account for volume fraction transport, the High-Resolution Interface Capturing scheme is specifically implemented to handle sharp phase interfaces. Free-surface flows with sharp interfaces present a discontinuity in the solution domain, causing discretisation errors. This can lead to spurious or parasitic currents. [18] Although often negligible, these currents can be accommodated for through the incorporation of an artificial viscosity term.

At large scale interfaces, the near-interface flow often exhibits laminar characteristics with the development of boundary layers. However, in situations with significant velocity gradients, turbulent fluctuations can arise in the regions adjacent to the interface (especially when the interface is mobile or highly distorted). Recognizing that standard RANS turbulence models tend to overpredict turbulence intensity near phase boundaries, STAR-CCM+ allows the user to apply turbulence damping strategies (e.g., vorticity limiters) which suppress excessive eddy viscosity and restore physically consistent laminar sublayers adjacent to the interface.

1.5 Phase Interactions

Phase interactions in free-surface flows and flash evaporation dictate how mass, momentum and energy are exchanged between contiguous liquid and gaseous domains. For both separated (e.g., stratified) and dispersed (e.g., bubbly) flow regimes, major phase interactions include:

- Drag force: Governs momentum exchange due to the relative motion between phases; typically decomposed using empirical drag laws (e.g., Schiller-Naumann for bubbles/droplets).
- Lift force: Acts perpendicular to the relative flow, important in shear flows with small dispersed phases; affects lateral migration of bubbles/droplets—uses models like Tomiyama, Saffman, or others depending on flow characteristics.
- Virtual mass force (added mass) Represents resistance to acceleration caused by the acceleration of surrounding fluid (significant in cases with rapid acceleration or deceleration of bubbles).
- Wall lubrication force Repels dispersed phase from solid walls, preventing unrealistic accumulation at boundaries (important in vertical/horizontal pipe flows).
- Turbulent dispersion force Spreads the dispersed phase due to turbulence in the continuous phase, modeled typically as a diffusion term proportional to turbulent kinetic energy gradient. The Favre-Averaged Drag (FAD) model approximates turbulent dispersion as a function of liquid eddy viscosity and bubble slip. This mitigates excessive bubble clustering below the free surface. [19]
- Surface tension Maintains interface shape, forms basis for capillary effects and controls droplet/bubble formation and coalescence/breakup.
- Interphase mass transfer (evaporation/condensation) Represents phase change, modeled as interfacial flux determined by local thermodynamic properties (temperature, concentration, pressure—often using kinetic models for flash evaporation).
- Interphase energy transfer Encompasses latent heat and sensible heat exchange at the interface and throughout the phases (essential for accurate phase change modeling).
- Nucleation (wall and bulk)
- Interfacial area changes due to bubble growth, coalescence, breakup
- Local pressure and temperature gradients driving simultaneous mass and energy transfer

1.6 Numerical Methods

- gradients
- solvers - implicit unsteady, segregated flow (SIMPLE)
- Relevant non-dim numbers? (reynolds, froude, jakobs, nusselt, bond, what not)

1.7 Workflow?

2 Subproblem: Liquid Infiltration

The first subproblem deals with water infiltrating the evaporation chamber during rocket ascent. The flow system assembly comprises of the hydraulic accumulator, the evaporation chamber and the injection system subassembly that drives the liquid from the hydraulic accumulator into the evaporation chamber. The hydraulic accumulator and the evaporation chamber can be heated until lift-off and current design considerations assume the temperature of the evaporation chamber and water in the accumulator to be $368.15K$ and $353.15K$ respectively. The ambient temperature at lift-off is assummed to be $280K$ based on the remarks and considerations made in [OLD-SED]. Although insulation from the environment outside the rocket is assumed and the hydraulic accumulator will be sufficiently insulated, the evaporation chamber can cool down slightly during ascent until the start of the experiment at time, $t = 140s$, after lift-off.

The over-pressure in the hydraulic accumulator that drives the water and the absolute pressure in the evaporation chamber are assumed to be $3.5bar$ and $1bar$ respectively. Water injection stops shortly before the apogee is reached and the flow is cut-off when the absolute pressure in the evaporation chamber rises to $1.7bar$. As no flow measurements are made in the injection assembly, and the hydraulic accumulator only contains a temperature sensor, it becomes necessary to estimate the inflow conditions of the water into the evaporation chamber. Using the pressure gradient between the accumulator and the evaporation chamber, the Darcy-Weisbach equation to estimate the head loss in the pipe and accomodating for the minor head losses incurred through the following components of the injection subassembly:

- adapter to reduce the pipe diameter from D to d
- two sharp $90deg$ connectors
- and six threaded connections in the piping,

the total head loss from the accumulator to the evaporation chamber can be written as,

$$h = (1 - \frac{d^2}{D^2})\frac{v^2}{2} + 2(1.1)\frac{v^2}{2} + 6(0.08)\frac{v^2}{2} + f\frac{L}{d}\frac{v^2}{2}, \tag{2.1}$$

where L is the length of the pipe, f is the Darcy friction factor, and v is the inflow velocity of the water into the evaporation chamber. The Darcy friction factor is determined using the Haaland approximation of the Colebrook equation and the velocity is estimated by solving the energy balance for the flow system assembly,

$$\frac{\Delta P}{\rho g} = \Delta h + \frac{v}{2} + h \tag{2.2}$$

The inflow velocity, v, is found to be. With a corresponding Reynolds number, $Re =$, the initial assumption of turbulent flow is validated.

With an initial pressure of $0.03MPa$ in the evaporation chamber, it is also necessary to validate if the continuum hypothesis remains valid for the air present in the chamber. The Knudsen number relates the mean free path, λ, to the scale length of the macroscopic flow property gradients and helps determine the validity of the continuum hypothesis. The magnitude of the mean free path can be used to estimate the Knudsen number.

$$\lambda = \frac{\mu}{\rho}\sqrt{\frac{\pi m}{2k_B T}} \tag{2.3}$$

For dry air, at a pressure and temperature of $0.03MPa$ and 328 respectively, molecular mass $m = 4.815e - 26$ kg, and estimating the kinematic viscosity using Sutherland's formula, the mean free path, $\lambda \approx 2.0732e - 06m$.

2.1 Objective

The objective of the first task is to model the water injection into the evaporation chamber and extract quantities of interest in order to better estimate the initial conditions for the second task. The validity of the multiphase physics models provided by Star-CCM+ and the equations implemented therein is also evaluated against the one-dimensional Stefan problem and the two-dimensional Rayleigh-Taylor instability.

2.2 Validation Problems

2.2.1 Stefan Problem

Phase changes can be regarded as the growth of the domain of one phase accompanied by the reduction of the domain of the other phase. This growth or reduction in domain sizes can be triggered by thermal as well as non-thermal processes [shumon koga] and this representative mathematical dynamic model containing moving boundaries is known as the Stefan system. The associated problem of analysing and finding the solutions to the aforementioned model is known as the Stefan problem. [koga]. The classical Stefan problem is a one dimensional solidification or melting problem citelaterjonas, jonsson. The one-dimensional Stefan problem under semi-infinite geometry, uniform initial temperature, constant boundary temperature, and thermophysical properties constant in each phase is probably the simplest, explicitly solvable problem of phase change [lectures, uni tenessee knoxville lectures], serving as a good benchmark to validate the energy equation implemented in the VOF solver in Star-CCM+. [DOES IT ALSO SERVE AS A GOOD BENCHMARK FOR SIMPLE HEAT CONDUCTION TOO??? BECOZ TWO BIRDS ONE STONE]

Assuming that the region in space, $\Omega \subset \mathbb{R}^2$, occupied by the material is the 2D semi-infinite rectangle $\Omega = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x \in [0,\infty), y \in [0,\Delta h]\}$, with the solid phase and the liquid phase occupying the regions Ω_s and Ω_l respectively such that $\Omega = \Omega_s \cup \Omega_l$. At any given time, t, the two phases are separated by the interface, $\Gamma(t)$, that marks the change in phase of the material. The boundary of the domain Ω is denoted by $\partial\Omega$. The problem space, $\mathcal{D}$, is the spatial and temporal domain occupied by the material, $\mathcal{D} = \{[t_0, t_f] \times \Omega\}$. With the heat equation governing the temperature field in the liquid region, the Stefan condition at the interface, and the respective boundary conditions, the system of equations defining the one-dimensional Stefan problem can be written as,

$$\begin{cases} \frac{\partial T}{\partial t} - \alpha \frac{\partial^2 T}{\partial x^2} & = 0, \\ \rho h_m \frac{d\Gamma(x(t),t)}{dt} + \kappa \frac{\partial \Gamma(x(t),t)}{\partial x} & = 0, \\ T(x,0) & = T_l, \\ \nabla T(x,t) \cdot \mathbf{n} & = 0, \\ T(x,0) & = T_m \end{cases} \tag{2.4}$$

- the system of equations is well-posed and has a unique solution. the evolution of the temperature field along with the evolution of the phase change interface is studied. The simulation assumes laminar and unsteady flow and solves for the solution variables implicitly using the SIMPLE scheme. The 2D region is discretised using a uniform orthogonal quadrilateral mesh of varying sizes in order to study the spatial convergence of the solution. The physical parameters for the simulation are selected according to water ice, as in [boledi et al]. A constant density $\rho = 1000\,\mathrm{kg \cdot m^{-3}}$, dynamic viscosity $\mu = 8.8871e-4\,\mathrm{Pa \cdot s}$, latent heat of fusion $100000\,\mathrm{J \cdot kg^{-1}}$, and specific heat $4200\,\mathrm{J \cdot kg^{-1}K^{-1}}$. Thermal conductivities are defined as $2.33\,\mathrm{W \cdot m^{-1}K^{-1}}$ and $0.62\,\mathrm{W \cdot m^{-1}K^{-1}}$ for the liquid and solid phases respectively. All four boundaries are assigned a no-slip shear stress specification. The boundary at $x = 0$ is assigned a constant temperature, $T = 273\,\mathrm{K}$, while the other boundaries are assumed to be adiabatic. The convection and diffusion terms are discretised using the second-order upwind scheme and the least squares method was used to interpolate the values of the gradients from the cell centres onto the face centres. The transient term is discretised using the implicit Euler-method which is first-order accurate. First-order HRIC is used to reconstruct the interface. Since Star-CCM+ sets under- and over-relaxation factors conservatively, the values are left unchanged.

The evolution of the temperature field for the simulation with cell size, $\Delta h = 5e-3$ is shown in Figure [FIG]. A clear propagation of the interface from the left to right is observed with increasing simulation times. The temperature also gradually decreases in the liquid phase from the Dirichlet boundary at $x = 0$ towards the melting temperature, T_m, at the interface after which it stays constant.

- 'This is exactly what one would expect for a velocity that is decreasing in time as is the case for the classical Stefan problem. One can also see that all simulation results converge to the analytical solution for a finer mesh and a smaller time step.'
- [ABOUT CONVERGENCE]
- [ABOUT COMPARISON OF TEMPERATURE W ANALYTICAL SOLUTION]
- [ABOUT COMPARISON OF PCI W ANALYTICAL SOLUTION]

2.2.2 Rayleigh-Taylor Instability

The Rayleigh-Taylor (RT) instability describes the growth of interfacial perturbations between two superposed fluids of differing densities when the heavier fluid overlies the lighter one under gravity [20]. This phenomenon, known as the Rayleigh-Taylor (RT) instability, is fundamental to many natural and engineering systems. Although the ideal, undisturbed RT instability is rarely encountered in practice, its essential features arise across a broad range of spatial and temporal scales—where the basic mechanism can be influenced by viscosity, surface tension, compressibility, external accelerations, or stratification[21] Perturbation growth is driven by oppositely directed gradients of pressure and density at the interface.

The density contrast between the two fluids is represented by the non-dimensional *Atwood number*, $At = (\rho_H - \rho_L)/(\rho_H + \rho_L)$, where ρ_H and ρ_L denote the densities of the upper (heavy) and lower (light) fluids, respectively. The Atwood number quantifies the strength of the instability, with larger values corresponding to greater density contrasts. The nonlinear development of the RT instability is central to a wide class of interfacial flows where one fluid penetrates or displaces another [22]. Accurately simulating RT instability serves as a classical benchmark for two-phase flow solvers, requiring detailed computation of interface dynamics and rigorous implementation of interphase interaction terms as discussed in [chapter stuf], and is widely used to validate numerical methods for multiphase flows [23].

We assume that the two immiscible, viscuous, and incompressible fluids occupy the regions,

$$\Omega_i(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : (-1)^i (y - h(t,x)) > 0,\ t \geq 0\},\ i = 1,2$$

and are separated by the interface, $\Gamma(t)$ where

$$\Gamma(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : y = h(t,x),\ t \geq 0\}$$

The flow is governed by the incompressible Navier-Stokes equations with variable density and viscosity fields according to the local phase distribution. Interface dynamics are subject to jump conditions expressed as

$$\begin{cases} \rho \frac{\partial(\mathbf{u})}{\partial t} + \rho \nabla \cdot (\mathbf{u}\mathbf{u}) & = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{g}\rho, \\ \nabla \cdot \mathbf{u} & = 0, \\ [\![S(u, p + \mathbf{g}\rho)\mathbf{n}]\!] & = \sigma\kappa\mathbf{n} + [\![\rho]\!]\mathbf{g}y \\ [\![u]\!] & = 0 \\ V & = \mathbf{u} \cdot \mathbf{n} \\ u(0) & = u_0, \\ \Gamma(0) & = \Gamma_0 \end{cases} \tag{2.5}$$

where ρ is the density, $\mathbf{u}$ is the velocity vector, p is the pressure, μ is the dynamic viscosity, and g is the gravitational acceleration.

The numerical simulations assume turbulent, unsteady flow and are performed using the SIMPLE implicit algorithm. The computational domain is a two-dimensional rectangle, $[0, L_x] \times [0, 4L_x]$, uniformly discretized by orthogonal quadrilateral cells. Convective and diffusive terms are discretized by the second-order upwind

scheme; cell-center gradients are interpolated to face centers using the least-squares method. Interface capturing is performed with the volume-of-fluid (VOF) method using the first-order HRIC scheme. The transient term is approximated using the first-order implicit Euler method. Turbulence effects are modeled using the Shear Stress Transport (SST) k-ω [better word?] closure, and the surface tension force is treated semi-implicitly. Initial conditions impose a single-mode interface perturbation, with amplitude, $A = 0.1L_x$, and wavelength, $\lambda = L_x$.

$$\Gamma_0 = 2L_x + A\cos\left(\frac{2\pi}{L_x}\right)$$

Gravitational acceleration acts downward, $g = 1.0$. Periodic boundary conditions are imposed in the horizontal direction and slip walls at the top and bottom. The Atwood number is varied, with $At \geq 0.5$ denoting the "high Atwood" regime [24], and physical parameters are chosen for dynamic similarity to canonical benchmark cases [25].

According to [18], a reasonable solution to the RT instability with $At \approx 0.75$ can be obtained with a 128×512 mesh. With similar mesh resolutions being used in [25, 26] to study similar RT instabilities, in addition to a mesh convergence study being performed, the domain is discretised using a uniform orthogonal quadrilateral mesh with size $\delta_h = 0.0045$. The timestep for the simulations is selected as $\delta t = 2.5 \times 10e-4/\sqrt{At}$ similar to [27, 25]. The following cases are studied:

Case	At	Re	σ	δ_h
1	0.5	300	$1e-5$	0.05
2	0.5	300	$1e-5$	0.007
3	0.5	300	$1e-5$	0.0045
4	0.5	300	$1e-5$	0.0025
5	0.2	1000	$1e-5$	0.0045
6	0.5	1000	$1e-5$	0.0045
5	0.5	3000	$1e-5$	0.0045
6	0.5	3000	$1e-5$	0.0045

Figure **??** illustrates the parameter definitions and nomenclature of an evolving single-mode [need i define single mode rt?] RT instability interface. The evolution of RT instabilities for Reynolds numbers 300, 1000, and 3000 are depicted in figure **??**. The initial perturbations, generally studied using linear stability analysis, experience an exponential growth [28]. When perturbations reach finite amplitude, nonlinear effects lead to the formation of rising "bubbles" of the lighter fluid and descending "spikes" of the heavier fluid. Continued evolution is marked by vortex formation and interface roll-up driven by shear and Kelvin-Helmholtz type instabilities [25, 22], which go on to contribute to turbulent mixing and chaotic flow features [21].

Figure [name figure] depicts the evolution of the interface at different times for varying density ratios as simulated in Star-CCM+. Upto high Reynolds numbers, the interface structure and its temporal evolution agree qualitatively with the results published in [25, 27, 26]. [why does it look different at high Re]. Using the bubble and spike heights, defined as the maximum vertical distance of the spike and bubble as measured from the undisturbed initial interface, the results can be quantitatively compared to [25] as shown in Figure **??**. Table **??** summarises the root mean square errors and the maximum errors between the simulated results and [25].

Simulations conducted for Atwood numbers very close to one also qualitatively show a similar progression and evolution of structures as the simulations in [26]. [What happens at high atwood numbers]

[Write about mesh convergence study. WHY DO RESULTS NOT MATCH?].

- [ABOUT SPATIAL AND TEMPORAL CONVERGENCE]

- [ABOUT EVOLUTION OF INTERFACE W VARYING REYNOLDS AND VARYING ATWOOD:] Figure [FIGFIG] and figure [GIFGIF] depict the evolution of the interface at different times for varying density ratios and Reynolds numbers respectively as simulated in Star-CCM+. Upto high Reynolds numbers, the temporal evolution and the structural features of the interface agree qualitatively with the results published in [hamzelhoom huang, ding]. As the solution evolves, the penetration of the spike increases and it eventually rolls up and forms two transient counter-rotating vortices due to shear forces and Kelvin-Helmholtz type instabilities [hamzelhoo], which go on to contribute to turbulent mixing and chaotic flow features [Sharp]. [WHY DO HIGH Re NUM SIMS DEVIATE FROM EXPECTED. MENTION SALIENT FEATURES OF THE EVOLUTION] Simulations conducted for Atwood numbers very close to one also qualitatively show a similar progression and evolution of structures as the simulations in [huang]. [What happens at high atwood numbers]
- [ABOUT COMPARISON OF BUBBLE HEIGHTS W DNS DATA AND L2 AND L-INF ERRORS:] Using the bubble and spike heights, defined as the maximum vertical distance of the spike and bubble as measured from the undisturbed initial interface, the results can be quantitatively compared to [hamzelhoo] as shown in Figure **??**. Table **??** summarises the root mean square errors and the maximum errors between the simulated results and [hamzelhoo].

2.2.3 Liquid Infiltration of Evaporation Chamber

- [ABOUT CONVERGENCE]
- [ON EVOLUTION OF INTERFACE and FLOW FIELD and RESULTING HEIGHT OF WATER]
- [ON EVOLUTION OF TEMPERATURE FIELD]

2.3 Conclusion?

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